SRN								

REVA UNIVERSITY
Bengaluru, India

REVA Academy for Corporate Excellence

Program: M. Tech/MSc in Artificial Intelligence	Course: Machine Learning
Batch: AI 09	Exam: Module End Exam
Date: 03-May-2025	Time: 10:00am to 1:00pm
Duration: 3 Hrs	Marks: 100

Instructions	
1.	This is an open-book exam. However, all the answers must be paraphrased and customised. Any evidence of copying verbatim from any third-party resources or plagiarised content will lead to suspension and other stringent actions. If any identical answers are found, both will be awarded zero.
2.	You may use any tool or software to answer the questions. All the outputs and codebase must be well commented.
3.	All the answers must be written in the answer script template provided by RACE. The template will be available in the LMS with the answer submission link. Along with the main answer script, all other artifacts like screenshots, codebase and any other materials must be submitted. All such artefacts must be named with the corresponding question number.
4.	All the artefacts must be zipped and submitted through the LMS Link. Zipped Folder must be named as 'SRN_Name of the Course'.

Instructions		Answer **ONE FULL** question from each unit.	**Marks**	
		UNIT - I		
Q1.	a.	Explain each stage of the machine learning lifecycle with the help of a real-world example.	5 marks	**25**
	b.	Explain, Naïve Bayes classifier based on the use case. In a dataset of emails, 10% are labeled as spam and 90% as not spam. 16% of emails contains the word "offer". The word "offer" appears in 80% of spam emails. Write its advantages and Disadvantages.	10 marks	
	c.	Define Principal Component Analysis (PCA) and its main objectives. Explain the steps involved in PCA. Discuss how you would decide the number of principal components to retain.	10 marks	
OR				
Q2.	a.	The following table shows information about patients, including their BMI, Exercise Level, and whether they have a certain disease. Using KNN algorithm with an appropriate value of k, determine the predicted result for the 6^{th} patient. Clearly show the steps involved, including distance calculations and final classification with justification.	10 marks	

S. No	BMI	Exercise Level	Disease
1	8	2	Yes
2	5	7	No
3	6	6	No
4	7	3	Yes
5	4	8	No
6	6	4	?

	b.	Describe and analyse the steps involved in training a Linear Regression model. Include the mathematical formulation, optimization process, and how model performance is evaluated using appropriate metrics.	10 marks	
	c.	Analyze the role of the sigmoid activation function in Logistic Regression and explain how it helps in mapping predictions to probabilities.	5 marks	
UNIT - II				
Q3.	a.	Explain the differences between Random Forest and Gradient Boosting in the context of ensemble learning.	5 marks	**25**
	b.	You have been given car dataset. Build the Random Forest and Gradient boosting models and compare the results. (No need to do extensive exploratory Data analysis).	10 marks	
	c.	Identify the top 5 important features and build the Random Forest model again and compare the results with the previous step. Briefly analyze the impact of feature reduction on model performance.	10 marks	
	OR			
Q4.	a.	You have developed a classification model to predict whether a customer will churn (leave) or not churn (stay). In your dataset, only 12% of customers actually churn, making it highly imbalanced. After testing the model, the following confusion matrix is obtained.	9 marks	

	Predicted Churn	Predicted Not churn
Actual Churn	30	90
Actual Not churn	40	840

i. Calculate Accuracy, Precision, Recall, and F1-Score.

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		ii. Is Accuracy the right metric to evaluate the model in this case? Justify your answer. iii. If your goal is to minimize customer loss by launching a retention campaign, should you prioritize Precision or Recall? Justify.		
	b.	What is K-Fold Cross Validation and how it evaluates model performance. Discuss its advantages and disadvantages.	**8 marks**	
	c.	What is overfitting in Machine learning? What measures can be taken to address overfitting in a Decision Tree model?	**8 marks**	
	UNIT - III			
Q5.	a.	Explain the steps of the K-Means Clustering algorithm. How do you choose the right number of clusters and write its limitations?	**10 marks**	25
	b.	You are given a bank dataset containing the following features: • DD (Demand Drafts) • Withdrawals • Deposits • Branch Area (in sq. ft.) • Average Daily Walk-ins Perform K-Means clustering on the given dataset by selecting relevant features, appropriately preprocessing the data (including handling missing values and scaling), determining the optimal number of clusters using the Elbow Method, and then applying the K-Means clustering algorithm based on the identified number of clusters.	**10 marks**	
	c.	Interpret the resulting clusters from question (b) in terms of bank operations.	**5 marks**	
	OR			
Q6.	a.	Explain the concept of the Silhouette Score used in clustering. Describe how it is calculated, what its value indicates about the quality of clustering, and how it can be used to determine the optimal number of clusters.	**10 marks**	25
	b.	Explain the different linkage methods used in hierarchical clustering—Single, Complete, Average, and Centroid Linkage. How does each method compute	**10 marks**	

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		the distance between clusters, and what are their typical use-cases, advantages, and limitations?		
	c.	What is a dendrogram in hierarchical clustering? Explain how it is constructed and how it helps in deciding the number of clusters.	**5 marks**	
	UNIT - IV			
Q7.	a.	Discuss the components of a time series: trend, seasonality, and noise. Provide suitable real-world examples to illustrate each component.	**6 marks**	**25**
	b.	You are given monthly airline passengers data (assume a clear seasonal pattern). Build three forecasting models: Holt-Winters Exponential Smoothing, ARIMA, and SARIMAX. Clearly outline your steps, assumptions, and key parameters used in each model.	**14 marks**	
	c.	Compare the three models using appropriate evaluation metrics using RMSE. Summarize your observations and comment on which model performs best and why?	**5 marks**	
	OR			
Q8.	a.	Consider the following dataset showing patient record (each row represents the symptoms and diagnosis observed in one patient). **Patient ID** — **Observed Items** 1 — Fever, Cough, Flu 2 — Fever, Flu 3 — Fever, Headache 4 — Cough, Flu 5 — Fever, Flu, Headache i. For the rule Fever → Flu. calculate Support, Confidence and lift. ii. Interpret the rule Fever → Flu based on the support, confidence, and lift values. Is this rule useful for early diagnosis? Justify.	**6 marks**	**25**
	b.	You are provided with an online retail dataset that contains transaction-level data, including InvoiceNo, Description, Quantity, and InvoiceDate.	**10 marks**	

Nested table from Q8 a.:

Patient ID	Observed Items
1	Fever, Cough, Flu
2	Fever, Flu
3	Fever, Headache
4	Cough, Flu
5	Fever, Flu, Headache

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		Using the Apriori algorithm, perform a market basket analysis on this dataset under the following constraints: • Minimum Support: 0.2 • Minimum Confidence: 0.8 • Minimum Lift: 6.0		
	c.	Based on the results from question(b), provide a clear interpretation of the discovered association rules and explain how they can be used to derive actionable insights in a retail business scenario.	**9 marks**	

Attached:

1. airline-passengers.csv
2. Carprice_dataset
3. online_retail_qp.xlsx

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